

Optimal-Restriction Five-Sets and Global Cut Stability in Triangle-Free Graph Bipartization

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Abstract

For a graph G , let

$$d(G) = |E(G)| - \text{MaxCut}(G)$$

be the minimum number of edges whose deletion makes G bipartite. A classical conjecture of Erdős asserts that every triangle-free graph on N vertices satisfies $d(G) \leq N^2/25$; for $N = 5k$ this is $d(G) \leq k^2$.

We investigate a five-vertex induction target

$$\exists Y \subseteq V(G), |Y| = 5 : \quad d(G) - d(G - Y) \leq 2k - 1.$$

Earlier versions of this project emphasized local recoloring certificates. The present note shifts to a global cut-stability phenomenon suggested by exact computations. For a maximum cut f of G , let $R_f(Y)$ be the number of monochromatic edges under f incident with Y , and let $\Delta_f(Y)$ be the additional decrease in monochromatic edges obtained by reoptimizing the cut on $G - Y$. Then exactly

$$d(G) - d(G - Y) = R_f(Y) + \Delta_f(Y).$$

We call Y *optimal-restriction preserving* if some maximum cut of G restricts to a maximum cut of $G - Y$, equivalently if $\Delta_f(Y) = 0$ for some $f \in \text{Opt}(G)$.

On a saved canonical corpus of 19,270 triangle-free graphs on 15 vertices, an exact full-cut audit finds that every saved graph admits a five-set Y with $d(G) - d(G - Y) \leq 5$ and with optimal-restriction preservation. The same conclusion holds for all 5,337 saved graphs with $L(G) \geq 12$, all 2,103 saved graphs with $d(G) > 5$, and all seven exactly identified edge-critical graphs. No failure occurs in the saved corpus. A later optimal-face diagnostic finds at least 218 low-defect preserving five-sets per saved graph.

We prove exact equivalences between optimal restriction, an intersection of optimal-cut faces under deletion, and a five-vertex extension frontier. Stronger shape properties of that frontier—monotonicity, one-step Lipschitz behavior, and discrete convexity—are explicitly falsified by saved witnesses. Thus the surviving phenomenon is genuinely global: for some five-set, the combined remainder-plus-extension cost is minimized at an optimal remainder coloring even though the extension frontier itself can oscillate strongly.

The full Erdős conjecture remains open. The contribution here is a reproducible global-cut phenomenon, exact equivalences that isolate it, and a sharply defined selection problem suggested by the full saved $n = 15$ corpus.

1 Introduction

For a finite simple graph G , define

$$d(G) = \min\{|F| : F \subseteq E(G), G - F \text{ is bipartite}\} = |E(G)| - \text{MaxCut}(G).$$

Erdős conjectured that every triangle-free graph on N vertices satisfies

$$d(G) \leq \frac{N^2}{25}.$$

The balanced blow-up of C_5 shows that $1/25$ would be sharp. Balogh, Clemen and Lidický proved the global $N^2/23.5$ bound and the sharp $N^2/25$ bound in two density ranges [1]. Ferudun gave a computer-assisted proof of the sharp value for multiples of five through $N = 200$ [2]. The computations below are therefore used for structure discovery and exact diagnostics, not as new finite extremal evaluations.

The induction target motivating this project is the following.

Conjecture 1 (Five-set induction target). *For every triangle-free graph G on $5k$ vertices, there exists a five-set Y such that*

$$d(G) - d(G - Y) \leq 2k - 1.$$

If this holds for every k , then induction gives

$$d(G) \leq 1 + 3 + \cdots + (2k - 1) = k^2.$$

The challenge is that deleting Y may cause the remaining graph to choose a substantially better global cut. This note isolates that reoptimization term and studies an empirical phenomenon in which it vanishes for a suitable five-set.

2 Corrected automatic threshold

For a five-set Y , define

$$\ell(Y) = 2d(G[Y]) + e(Y, G - Y), \quad L(G) = \min_{|Y|=5} \ell(Y).$$

The coordinated-flip selection inequality from the earlier project gives

$$d(G) - d(G - Y) \leq \left\lfloor d(G[Y]) + \frac{e(Y, G - Y)}{2} - \Phi_Y \right\rfloor,$$

where $\Phi_Y \geq 0$ is an averaged recoloring gain. Taking the empty flip immediately yields

$$d(G) - d(G - Y) \leq \left\lfloor \frac{\ell(Y)}{2} \right\rfloor.$$

Hence:

Proposition 1 (Automatic threshold). *If G has $5k$ vertices and some five-set Y satisfies*

$$\ell(Y) \leq 4k - 1,$$

then

$$d(G) - d(G - Y) \leq 2k - 1.$$

At $k = 3$, the automatic range extends through $L(G) = 11$. This corrects the threshold used in Version 1 of the project. Consequently the older $L = 11$ exchange audit remains a valid structural computation but is not needed to establish the induction inequality in that regime.

3 Global reoptimization after deleting five vertices

Fix a maximum cut f of G , represented as a 2-coloring. For a five-set Y , put $H = G - Y$ and let $\text{mono}_f(H)$ denote the number of monochromatic edges of H under the restriction of f .

Define

$$R_f(Y) = d(G) - \text{mono}_f(H),$$

which is exactly the number of monochromatic edges of f having at least one endpoint in Y . Define the remainder reoptimization gain

$$\Delta_f(Y) = \text{mono}_f(H) - d(H) \geq 0.$$

Then:

Lemma 1 (Exact reoptimization decomposition). *For every maximum cut f of G and every five-set Y ,*

$$d(G) - d(G - Y) = R_f(Y) + \Delta_f(Y).$$

Proof. Since f is globally optimal, $d(G)$ is its monochromatic-edge count. Partition those monochromatic edges into the ones incident with Y and the ones wholly inside H . The latter number is $\text{mono}_f(H) = d(H) + \Delta_f(Y)$. $\square$

The local term $R_f(Y)$ is easy to average. The difficulty is the global term $\Delta_f(Y)$, which measures how much the entire remainder can reorganize after deletion.

4 Optimal-restriction five-sets

Definition 1. *A pair (f, Y) is optimal-restriction preserving if $f \in \text{Opt}(G)$ and the restriction $f|_{G-Y}$ is optimal for $G - Y$. Equivalently,*

$$\Delta_f(Y) = 0.$$

We call Y an H0 witness if it is good for the induction bound and is optimal-restriction preserving for at least one maximum cut:

$$d(G) - d(G - Y) \leq 2k - 1, \quad \exists f \in \text{Opt}(G) : \Delta_f(Y) = 0.$$

For such a pair,

$$d(G) - d(G - Y) = R_f(Y),$$

so the global reoptimization difficulty disappears completely.

The following exact audit motivated the rest of the note.

Proposition 2 (Saved-corpus H0 phenomenon). *In the saved canonical $n = 15$ corpus used by this project, every one of 19,270 graphs has an H0 witness. In particular, the property holds for all 5,337 saved graphs with $L(G) \geq 12$, all 2,103 saved graphs with $d(G) > 5$, and all seven exactly identified edge-critical graphs. No failure occurs in the saved corpus.*

This proposition is a statement about the precisely defined saved corpus. It is not claimed that the saved corpus exhausts all triangle-free graphs on 15 vertices.

A representative-sample audit first suggested the phenomenon on 40 graphs. The exact full-corpus run then verified all 19,270 saved graphs, with independent recomputation of the optimum of every saved witness core. Both

$$\max_G \min_{Y \text{ good}} \Delta_{\min}(Y) = 0$$

and the corresponding minimum cut-distance statistic were zero.

5 Abundance rather than uniqueness

The H0 witness is not rare in the saved data. Subsequent diagnostics found that every saved graph has at least 218 low-defect five-sets whose optimal-cut restriction family meets the optimal-cut family of the remainder. A separate maximum-cut-oriented diagnostic found many such witnesses per optimum cut in the saved corpus.

However, abundance alone does not prove the desired theorem. Even for the balanced blow-up B_3 , conditioning on preservation does not necessarily lower the mean defect count below the target. Thus a proof requires more than a lower bound on the number of preserving five-sets.

6 Optimal-face projection formulation

For a five-set Y , define

$$\text{Res}_Y(G) = \{f|_{G-Y} : f \in \text{Opt}(G)\}.$$

Then the H0 preservation property is exactly an intersection statement.

Lemma 2 (Optimal-face equivalence). *For a fixed five-set Y , the following are equivalent:*

1. *there exists $f \in \text{Opt}(G)$ with $f|_{G-Y} \in \text{Opt}(G - Y)$;*
2. $\text{Res}_Y(G) \cap \text{Opt}(G - Y) \neq \emptyset$;
3. *there exists $f \in \text{Opt}(G)$ with $\Delta_f(Y) = 0$.*

The full saved-corpus diagnostics verify that every saved graph has many low-defect five-sets with nonempty intersection. A direct projection counting identity can be obtained by double counting pairs (f, Y) , but naive averaging is obstructed because distinct remainder optima can have very different numbers of extensions back to global optima. Thus the remaining proof problem is weighted rather than purely cardinality based.

7 The extension frontier

The same phenomenon has a second exact formulation that makes the remainder/extension tradeoff explicit. Fix Y and put $H = G - Y$. For each attainable integer $j \geq 0$, define

$$F_Y(j) = \min\{\text{extension cost over } Y : \text{the coloring of } H \text{ has } d(H) + j \text{ monochromatic edges}\},$$

where the extension cost counts monochromatic edges in $G[Y]$ and between Y and H , minimized over the 2^5 assignments on Y .

Lemma 3 (Extension-frontier identity). *For every five-set Y ,*

$$d(G) - d(G - Y) = \min_{j \geq 0} (j + F_Y(j)).$$

Moreover, Y is optimal-restriction preserving if and only if the minimum is attained at $j = 0$, equivalently

$$F_Y(0) = d(G) - d(G - Y).$$

Thus the exact H0 condition is

$$j + F_Y(j) \geq F_Y(0) \quad \text{for every attainable } j \geq 1.$$

The saved H0 witnesses show that this anchored combined-cost inequality is robust even though much stronger frontier-shape hypotheses are false. Among all 19,270 saved witness frontiers, only 7,745 satisfy the tested adjacent finite-layer slope condition, and only 2,405 have F_Y itself minimized at zero. Saved witnesses can have large downward adjacent jumps. In particular, monotonicity, one-step Lipschitz behavior, and discrete convexity of F_Y are not universal.

The empirical statement that survives is weaker and more precise:

$$\text{for some useful } Y, \quad j + F_Y(j) \text{ is minimized at } j = 0.$$

8 Why several natural proof shortcuts fail

The computational program deliberately tested and rejected several stronger statements.

Fixed-set local certificates. Singleton, pair, star, biclique, and a 32-flip local system provide exact identities and useful conditional bounds, but purely local five-set data does not control global remainder reoptimization in general.

Simple witness selection rules. Feature mining over all saved five-sets found no universal simple rule such as minimizing degree sum, boundary size, or a fixed induced five-vertex type. A strong two-statistic rule covered 19,133/19,270 saved graphs but still failed on 137.

Hitting-set and uncrossing arguments. Minimal dangerous remainder flips can be taken connected, but can be large. The relevant signed cut-gain function is not generally submodular, so standard laminarity/uncrossing arguments do not directly force a five-vertex transversal.

Lexicographic gain profiles. Refining the objective by the multiplicities of gain levels does not force descent: isomorphic remainders can preserve the entire gain profile while replacing maximizing supports.

Naive abundance averaging. There may be many preserving deletions, yet the preserved-family conditional mean defect can exceed the target. The obstruction is an extension-weight bias: remainder optima can have different numbers of extensions to global optima.

These failures are useful diagnostics. They indicate that the remaining problem is genuinely a global selection statement about the relation between the optimal-cut families of G and $G - Y$.

9 Current theorem target

The full saved-corpus evidence suggests the following global stability statement.

Conjecture 2 (Optimal-restriction five-set principle). *Let G be triangle-free on $5k$ vertices. Under the structural hypotheses forced by a genuine minimal counterexample to the Erdős bound, there exists a five-set Y and a maximum cut f of G such that*

$$f|_{G-Y} \in \text{Opt}(G - Y)$$

and

$$R_f(Y) \leq 2k - 1.$$

Equivalently,

$$d(G) - d(G - Y) \leq 2k - 1$$

with no remainder reoptimization loss.

A proof would close the five-set induction route immediately. The current exact blocker is to show that not every low-defect deletion pair can fail optimal projection. In optimal-face language, one seeks positivity of the low-defect intersection count after correctly accounting for extension multiplicities. In frontier language, one seeks a five-set with $F_Y(0) \leq 2k - 1$ for which no positive remainder-cost layer beats $j = 0$.

10 Reproducibility and scope

The project contains exact bitmask MaxCut code, deterministic full-cut landscape audits, saved-corpus witness tables, independent validation tests, and SHA-256 checksums for release artifacts. The v3 global-cut analyses were performed on the saved canonical $n = 15$ corpus of 19,270 graphs. The principal new artifacts are the global-cut landscape summaries, H0 witness tables, extension-frontier diagnostics, and optimal-face projection notes.

The following distinctions are maintained throughout:

1. exact symbolic identities;
2. exact finite computations on explicitly defined saved corpora;
3. conjectural general statements motivated by those computations.

No claim is made that the saved 19,270-graph corpus is the complete universe of triangle-free graphs on 15 vertices unless a separate exhaustive-generation proof is supplied.

11 Discussion

The computational search changed direction substantially over the course of the project. Early work focused on increasingly rich local recoloring certificates. Exact counterexamples to several stronger local statements showed that the difficult part is not the five deleted vertices themselves, but the global reoptimization of the remainder.

The full saved-corpus H0 audit then exposed a cleaner global pattern: after choosing the right five-set, the troublesome reoptimization term vanishes. That phenomenon survived all 19,270 saved graphs and all tested hard subclasses, while many stronger regularity properties of the extension frontier failed. The current evidence therefore points toward a theorem about projection/intersection of optimal-cut families rather than a theorem about local flip geometry.

Status and non-claims

The Erdős triangle-free bipartization conjecture remains open. This manuscript does not claim a proof of the conjecture or of the five-set induction target in general. The 19,270-graph H0 statement is an exact saved-corpus result, not an exhaustive $n = 15$ classification claim. The purpose of the v3 note is to document the global cut-stability phenomenon, its exact equivalent formulations, and the strongest proof barriers identified so far.

References

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